

# Benchmarking Generative Models on ArtBench-10: A Comparative Study of VAEs, GANs, and DDPMs

João Vaz and Leonardo Silva

Department of Informatics Engineering, Coimbra, Portugal  
{joaoafvaz, leonardosilva}@student.dei.uc.pt

**Abstract.** We present a comparative study of three generative model families,  $\beta$ -VAE, DCGAN, and conditional DDPM with classifier-free guidance and DDIM sampling, trained on ArtBench-10, a benchmark of  $32 \times 32$  paintings including ten artistic styles. All models are evaluated under a controlled protocol using FID and KID reported as mean  $\pm$  standard deviation across ten random seeds, with the best-performing configuration subsequently retrained on the full dataset. The conditional DDPM achieves  $\text{FID} = 9.28 \pm 0.10$  on the full dataset, surpassing all unconditional diffusion baselines reported in the original study [10]. DCGANs offer competitive quality at lower computational cost, while  $\beta$ -VAEs are limited by their reconstruction objective.

**Keywords:** Generative Models · Image Synthesis · Variational Autoencoders · Generative Adversarial Networks · Denoising Diffusion Probabilistic Models · ArtBench.

## 1 Introduction

Generative models have become a central tool in computer vision and creative AI, enabling the synthesis of realistic and diverse images from learned data distributions. Their application to artistic domains is particularly compelling, as art exhibits rich structural patterns, stylistic coherence, and high perceptual complexity that challenge standard generative pipelines.

In this work, we study generative modeling on the ArtBench-10 dataset [10], a benchmark designed to evaluate generative models on artwork images grouped by artistic style. We implement and compare three prominent model families: Variational Autoencoders (VAEs) [9], which learn a structured latent space via probabilistic encoding, Deep Convolutional GANs [12], which employ an adversarial training objective between a generator and a discriminator, and Denoising Diffusion Probabilistic Models (DDPMs) [6] with Classifier-Free Guidance [7] and DDIM sampling [14], which learn to reverse a gradual noising process for conditional generation per artistic style.

## 2 Problem Description

ArtBench-10 consists of 60,000 images (50,000 for training and 10,000 for testing) at a  $32 \times 32$  resolution, distributed across ten distinct artistic styles. The challenge of generative modeling in this context lies in balancing visual fidelity, sample diversity, and stylistic consistency. To ensure a robust assessment, we follow a rigorous evaluation protocol: 5,000 generated samples are compared against 5,000 real images using Fréchet Inception Distance (FID) [4] and Kernel Inception Distance (KID) [1]. To account for statistical variability, all metrics are reported as the mean and standard deviation across ten independent random seeds.

## 3 Methodology and Approach

### 3.1 Variational Autoencoders

Variational Autoencoders (VAEs) [9] learn a probabilistic mapping from input images to a structured latent space. The encoder  $q_\phi(\mathbf{z} \mid \mathbf{x})$  produces a Gaussian posterior parameterised by mean  $\boldsymbol{\mu}$  and log-variance  $\log \boldsymbol{\sigma}^2$ ; the decoder  $p_\theta(\mathbf{x} \mid \mathbf{z})$  reconstructs the image from a sample  $\mathbf{z} \sim q_\phi$ . Training minimises the Evidence Lower Bound (ELBO):

$$\mathcal{L}_{\beta\text{-VAE}} = \mathbb{E}_{q_\phi} [\log p_\theta(\mathbf{x} \mid \mathbf{z})] - \beta D_{\text{KL}}(q_\phi(\mathbf{z} \mid \mathbf{x}) \parallel p(\mathbf{z})) \quad (1)$$

where  $\beta \geq 1$  controls the weight of the KL regularisation term relative to the reconstruction loss [5]. Values of  $\beta < 1$  prioritise reconstruction fidelity, whereas larger values impose a stronger disentanglement pressure on the latent space at the cost of generation quality.

**Architecture.** The architectural design of the encoder consists of three strided convolutional blocks ( $3 \rightarrow 64 \rightarrow 128 \rightarrow 256$  channels, kernel size 4, stride 2, padding 1), each followed by Batch Normalisation and LeakyReLU activations (slope 0.2). The resulting  $256 \times 4 \times 4$  feature map is flattened and projected to  $\boldsymbol{\mu}$  and  $\log \boldsymbol{\sigma}^2$  via separate linear layers. The decoder mirrors this structure: a linear layer projects the latent vector  $\mathbf{z}$  back to  $256 \times 4 \times 4$ , followed by three transposed convolutional blocks ( $256 \rightarrow 128 \rightarrow 64 \rightarrow 3$  channels) with Batch Normalisation and ReLU activations. A Sigmoid activation at the final layer maps outputs to  $[0, 1]$  to match the normalisation of the input images. Weights are initialised from  $\mathcal{N}(0, 0.02)$  for convolutional and linear layers, with Batch Normalisation scales set to 1 and biases to 0.

**Reparameterisation Trick.** To allow gradients to flow through the stochastic sampling step, latent samples are drawn as  $\mathbf{z} = \boldsymbol{\mu} + \boldsymbol{\sigma} \odot \boldsymbol{\epsilon}$ , where  $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ . At inference time,  $\boldsymbol{\epsilon}$  is omitted and the deterministic mean  $\boldsymbol{\mu}$  is used directly.

**Loss Function.** The reconstruction term is the mean squared error summed over pixels and normalised by batch size. The KL term is computed in closed form as:

$$D_{\text{KL}} = -\frac{1}{2} \sum_{j=1}^d (1 + \log \sigma_j^2 - \mu_j^2 - \sigma_j^2) \quad (2)$$

normalised by batch size. The  $\beta$  coefficient is annealed linearly from 0 to its target value over the first 60% of training epochs, stabilising early reconstruction before progressively enforcing the KL constraint.

**Training Details.** The model is optimised with Adam ( $\beta_1 = 0.9$ ,  $\beta_2 = 0.999$ ) at a learning rate of  $10^{-3}$ , decayed via cosine annealing to  $10^{-5}$ . Gradient norms are clipped at 1.0. The batch size is 64 and all models are trained for 300 epochs on the 20% subset. Images are normalised to  $[0, 1]$  using a `ToTensor` transform without additional augmentation.

### 3.2 Generative Adversarial Networks

Generative Adversarial Networks (GANs) [3] frame image synthesis as a two-player minimax game between a generator  $G$  and a discriminator  $D$ . The generator maps a latent vector  $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$  to a synthetic image, while the discriminator attempts to distinguish real images from generated ones. Training minimises the adversarial objective:

$$\min_G \max_D \mathbb{E}_{\mathbf{x} \sim p_{\text{data}}} [\log D(\mathbf{x})] + \mathbb{E}_{\mathbf{z} \sim p(\mathbf{z})} [\log(1 - D(G(\mathbf{z})))] \quad (3)$$

**Architecture.** We adopt the Deep Convolutional GAN (DCGAN) [12] architecture. The generator  $G$  maps a latent vector  $\mathbf{z} \in \mathbb{R}^{100}$  to a  $3 \times 32 \times 32$  image through a series of transposed convolutions, each followed by Batch Normalisation and ReLU activations, with a final Tanh activation to produce outputs in  $[-1, 1]$ . The discriminator  $D$  mirrors this structure with strided convolutions, LeakyReLU activations with slope 0.2, and Batch Normalisation in all layers except the first. The base number of feature maps is controlled by *ngf* for the generator and *ndf* for the discriminator.

**Training Objective.** Rather than the original binary cross-entropy loss, we use the numerically more stable Binary Cross-Entropy with Logits loss (BCEWithLogitsLoss), which fuses the sigmoid activation and the loss computation into a single operation. The discriminator is trained to classify real images as 1 and generated images as 0, while the generator is trained to fool the discriminator by maximising the discriminator score on generated samples.

**Label Smoothing.** To prevent the discriminator from becoming overconfident on real samples, we apply one-sided label smoothing [13], replacing the hard target of 1 for real images with a soft target of 0.9. This encourages the discriminator to maintain uncertainty, providing more informative gradients to the generator throughout training.

**Asymmetric Learning Rates.** To address the training imbalance observed when the discriminator dominates early in training, we apply an asymmetric learning rate schedule: the generator is trained at learning rate  $2 \times 10^{-4}$ , while the discriminator uses a reduced rate of  $1 \times 10^{-4}$ . This slows down discriminator learning relative to the generator, promoting a more balanced adversarial dynamic.

**Training Details.** Both networks are optimised with the Adam [8] optimiser with  $\beta_1 = 0.5$  and  $\beta_2 = 0.999$ , following the standard recommendation for GAN training [12]. Weights are initialised from  $\mathcal{N}(0, 0.02)$  for convolutional layers and  $\mathcal{N}(1, 0.02)$  for Batch Normalisation layers. Images are normalised to  $[-1, 1]$  and all models are trained for 300 epochs on the 20% subset.

### 3.3 Denoising Diffusion Probabilistic Models

Denoising Diffusion Probabilistic Models (DDPMs) [6] are a class of generative models that learn to reverse a gradual noising process. The forward process progressively corrupts a real image  $\mathbf{x}_0$  over  $T$  timesteps by adding Gaussian noise, until the signal is indistinguishable from pure noise  $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ . The reverse process learns to denoise  $\mathbf{x}_t$  back to  $\mathbf{x}_0$  by predicting the noise  $\epsilon$  added at each step.

**Forward Process.** The forward process is defined by a cosine noise schedule [11], which controls the variance of the noise added at each timestep. Given a real image  $\mathbf{x}_0$ , the noisy image at timestep  $t$  can be sampled directly as:

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, \quad \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I}) \quad (4)$$

where  $\bar{\alpha}_t$  is the cumulative product of noise coefficients at timestep  $t$ . The cosine schedule was adopted over the linear schedule as it avoids abrupt transitions at the extremes of the noising process, leading to more stable training [11].

**U-Net Architecture.** The reverse process is parameterised by a U-Net that receives the noisy image  $\mathbf{x}_t$ , the timestep  $t$ , and optionally the class label  $y$ , and predicts the noise  $\epsilon_\theta(\mathbf{x}_t, t, y)$ . The architecture consists of three resolution levels ( $32 \times 32$ ,  $16 \times 16$ ,  $8 \times 8$ ) with skip connections between encoder and decoder. Each level contains  $N$  residual blocks with GroupNorm and SiLU activations. Timestep information is injected via sinusoidal positional embeddings projected

into each residual block. A self-attention layer is applied at the  $8 \times 8$  bottleneck to capture long-range dependencies. The base channel width is  $C = 128$ , yielding channel dimensions of  $C$ ,  $2C$ , and  $4C$  across the three levels.

**Classifier-Free Guidance.** To enable conditional generation per artistic style, we adopt Classifier-Free Guidance (CFG) [7]. During training, class labels are randomly dropped with probability  $p_{\text{uncond}} = 0.10$ , replacing them with a null token. This forces the model to learn both the conditional distribution  $p(\mathbf{x}|y)$  and the unconditional distribution  $p(\mathbf{x})$  within a single network. At sampling time, the predicted noise is computed as:

$$\tilde{\epsilon} = \epsilon_{\theta}(\mathbf{x}_t, t) + w (\epsilon_{\theta}(\mathbf{x}_t, t, y) - \epsilon_{\theta}(\mathbf{x}_t, t)) \quad (5)$$

where  $w$  is the guidance scale. Higher values of  $w$  increase style fidelity at the cost of sample diversity.

**Sampling.** Rather than running the full  $T = 1000$  reverse steps, we adopt the Denoising Diffusion Implicit Models (DDIM) sampler [14], which allows high-quality generation in a fraction of the steps by reformulating the reverse process as a non-Markovian deterministic trajectory.

**Exponential Moving Average.** To stabilise sampling quality, we maintain an Exponential Moving Average (EMA) of the model weights throughout training with decay  $\rho = 0.9999$ . The EMA weights are used exclusively for evaluation and sampling, while the original weights continue to be updated by the optimiser. This prevents transient instabilities in the final training epochs from degrading generation quality.

**Training Details.** The model is trained with the AdamW optimiser at a learning rate of  $2 \times 10^{-4}$  with weight decay  $10^{-4}$  and gradient clipping at 1.0. A linear warmup of 5 epochs is applied at the start of training. The batch size is 128 and all models are trained for 500 epochs on the 20% subset unless the final one. Images are normalised to  $[-1, 1]$  and random horizontal flipping is applied as data augmentation.

## 4 Experimental Setup

All models are trained on a 20% subset for ablation and the best configuration retrained on the full dataset. For  $\beta$ -VAEs, we evaluate  $\beta \in \{0.1, 0.5, 1.0\}$  and  $d_z \in \{64, 128, 256\}$ . For DCGANs, we test label smoothing, asymmetric learning rates, and  $ngf \in \{64, 128, 256\}$ . For DDPMs, we evaluate guidance scale  $w \in \{1.5, 3.0, 5.0\}$ , capacity  $C \in \{64, 128\}$ , EMA, and training duration (300 vs. 500 epochs).

## 5 Analysis of Results

### 5.1 Variational Autoencoders

Table 1 reports FID and KID for all  $\beta$ -VAE configurations.

**Table 1.**  $\beta$ -VAE results on the 20% subset. FID and KID reported as mean  $\pm$  std over ten random seeds.

| Model               | $\beta$    | $d_z$     | FID $\downarrow$                   | KID $\downarrow$                        |
|---------------------|------------|-----------|------------------------------------|-----------------------------------------|
| <b>vae_b01</b>      | 0.1        | 128       | $98.29 \pm 0.49$                   | $0.11309 \pm 0.00136$                   |
| <b>vae_b05</b>      | 0.5        | 128       | $112.17 \pm 0.47$                  | $0.09603 \pm 0.00061$                   |
| <b>vae_b1</b>       | 1.0        | 128       | $136.65 \pm 0.65$                  | $0.11660 \pm 0.00164$                   |
| <b>vae_b01_z64</b>  | <b>0.1</b> | <b>64</b> | <b><math>97.39 \pm 0.38</math></b> | <b><math>0.10608 \pm 0.00123</math></b> |
| <b>vae_b01_z256</b> | 0.1        | 256       | $102.76 \pm 0.48$                  | $0.12231 \pm 0.00125$                   |

**Effect of the KL Regularisation Coefficient.** Increasing  $\beta$  consistently degrades FID: **vae\_b01** achieves  $\text{FID} = 98.29 \pm 0.49$ , while **vae\_b05** and **vae\_b1** reach  $112.17 \pm 0.47$  and  $136.65 \pm 0.65$ , respectively. This monotonic degradation reflects the well-known trade-off between regularisation and reconstruction quality in  $\beta$ -VAEs: stronger KL pressure forces the posterior towards the prior, compressing the latent space and reducing the decoder’s ability to faithfully reconstruct fine artistic texture. The KID metric exhibits a non-monotonic pattern, with **vae\_b05** achieving the lowest value (0.09603), suggesting that a moderate  $\beta$  may marginally improve the alignment between generated and real feature distributions even as overall sample fidelity degrades relative to **vae\_b01**.

**Effect of Latent Dimensionality.** Across the three latent sizes tested at  $\beta = 0.1$ , performance follows a monotonic trend where increasing the latent dimensionality  $d_z$  consistently degrades generative quality. Reducing  $d_z$  from 128 to 64 yields a slight improvement in FID ( $97.39 \pm 0.38$  vs  $98.29 \pm 0.49$ ) and a lower KID (0.10608 vs 0.11309), suggesting that the 64-dimensional bottleneck provides a mild regularising effect that benefits generation quality. In contrast, expanding to  $d_z = 256$  degrades both metrics ( $\text{FID} = 102.76 \pm 0.48$ ,  $\text{KID} = 0.12231 \pm 0.00125$ ), consistent with the posterior becoming harder to match to the standard Gaussian prior as capacity grows without a corresponding increase in the KL coefficient.

### 5.2 Generative Adversarial Networks

Table 2 reports FID and KID for all DCGAN configurations.

**Table 2.** DCGAN results on the 20% subset. FID and KID reported as mean  $\pm$  std over ten random seeds.

| Model                      | <i>ngf</i> | FID $\downarrow$                   | KID $\downarrow$                        |
|----------------------------|------------|------------------------------------|-----------------------------------------|
| <b>dcgan_base</b>          | 64         | $22.42 \pm 0.24$                   | $0.01465 \pm 0.00079$                   |
| <b>dcgan_ls</b>            | 64         | $22.04 \pm 0.42$                   | $0.01476 \pm 0.00067$                   |
| <b>dcgan_ls_lr2</b>        | 64         | $21.60 \pm 0.22$                   | $0.01378 \pm 0.00059$                   |
| <b>dcgan_ls_lr4</b>        | 64         | $26.46 \pm 0.34$                   | $0.01917 \pm 0.00079$                   |
| <b>dcgan_ls_lr2_ngf128</b> | 128        | $20.13 \pm 0.28$                   | $0.01272 \pm 0.00058$                   |
| <b>dcgan_ls_lr2_ngf256</b> | <b>256</b> | <b><math>18.79 \pm 0.36</math></b> | <b><math>0.01201 \pm 0.00053</math></b> |
| Real vs. Real              | —          | $2.51 \pm 0.03$                    | $-0.00003 \pm 0.00020$                  |

**Effect of Training Stabilisation.** Label smoothing alone produces a marginal improvement over the baseline, reducing FID from  $22.42 \pm 0.24$  to  $22.04 \pm 0.42$ . The combination of label smoothing with a halved discriminator learning rate (LR/2) yields a more consistent improvement, reaching FID =  $21.60 \pm 0.22$  and the lowest KID among the *ngf* = 64 configurations. This suggests that slowing down discriminator updates relative to the generator promotes a more balanced adversarial dynamic, consistent with the loss curves observed during training, where the discriminator loss stabilises around 0.6 rather than collapsing towards zero. Reducing the discriminator learning rate further (LR/4) proves detrimental, yielding FID =  $26.46 \pm 0.34$ , as the discriminator becomes too weak to provide informative gradients early in training, resulting in a prolonged and unstable convergence phase.

**Effect of Generator Capacity.** Increasing the base number of feature maps from *ngf* = 64 to *ngf* = 128 and *ngf* = 256 consistently improves generation quality. FID drops from  $21.60 \pm 0.22$  to  $20.13 \pm 0.28$  and  $18.79 \pm 0.36$  respectively, while KID follows the same trend. The improvement of approximately 1.5 FID points per capacity doubling suggests that the generator benefits from additional representational capacity to capture the stylistic diversity of ArtBench-10, though with diminishing returns. Visual inspection of generated samples confirms this trend: *ngf* = 256 produces noticeably sharper compositions and more recognisable artistic styles compared to *ngf* = 64.

### 5.3 Denoising Diffusion Probabilistic Models

Table 3 reports FID and KID for all DDPM configurations across training durations.

**Effect of Training Duration.** Extending training from 300 to 500 epochs produces a dramatic improvement for CFG-enabled models. **cfg\_w3** drops from FID =  $25.04 \pm 0.25$  at 300 epochs to FID =  $12.29 \pm 0.31$  at 500 epochs, a reduction of approximately 51%. A similar trend is observed for **cfg\_w2**, which improves from FID =  $28.04 \pm 0.30$  to  $12.79 \pm 0.24$ . In contrast, the unconditional baseline

**Table 3.** DDPM results on the 20% subset and full dataset. FID and KID reported as mean  $\pm$  std over ten random seeds.

| Model                            | Epochs     | FID $\downarrow$                  | KID $\downarrow$                        | DDIM steps |
|----------------------------------|------------|-----------------------------------|-----------------------------------------|------------|
| <b>baseline</b> (no CFG, no EMA) | 300        | $22.08 \pm 0.32$                  | $0.01590 \pm 0.00071$                   | 20         |
| <b>baseline</b> (no CFG, no EMA) | 500        | $21.97 \pm 0.37$                  | $0.01606 \pm 0.00073$                   | 20         |
| <b>cfg_w1</b> ( $w = 1.5$ )      | 300        | $33.36 \pm 0.28$                  | $0.02566 \pm 0.00070$                   | 20         |
| <b>cfg_w2</b> ( $w = 3.0$ )      | 300        | $28.04 \pm 0.30$                  | $0.01948 \pm 0.00066$                   | 20         |
| <b>cfg_w2</b> ( $w = 3.0$ )      | 500        | $12.79 \pm 0.24$                  | $0.00632 \pm 0.00043$                   | 20         |
| <b>cfg_w3</b> ( $w = 5.0$ )      | 300        | $25.04 \pm 0.25$                  | $0.01388 \pm 0.00049$                   | 20         |
| <b>cfg_w3</b> ( $w = 5.0$ )      | 500        | $12.29 \pm 0.31$                  | $0.00378 \pm 0.00044$                   | 20         |
| <b>cap_64</b> ( $C = 64$ )       | 500        | $15.19 \pm 0.24$                  | $0.00500 \pm 0.00041$                   | 20         |
| <b>cfg_w3</b> ( $C = 128$ )      | 500        | $12.29 \pm 0.31$                  | $0.00378 \pm 0.00044$                   | 20         |
| <b>no_ema</b>                    | 500        | $12.22 \pm 0.24$                  | $0.00483 \pm 0.00056$                   | 20         |
| <b>cfg_w3</b> (EMA)              | 500        | $12.29 \pm 0.31$                  | $0.00378 \pm 0.00044$                   | 20         |
| <b>best</b> (full dataset)       | 500        | $9.67 \pm 0.25$                   | $0.00406 \pm 0.00049$                   | 20         |
| <b>best (full dataset)</b>       | <b>500</b> | <b><math>9.28 \pm 0.10</math></b> | <b><math>0.00264 \pm 0.00031</math></b> | <b>50</b>  |
| Real vs. Real                    | -          | $2.51 \pm 0.03$                   | $-0.00003 \pm 0.00020$                  | -          |

shows negligible improvement between 300 and 500 epochs (22.08 vs. 21.97), this suggests that while unconditional structural features are learned early, the alignment between class embeddings and image features in conditional models requires a higher number of iterations. Consequently, the 300 epoch results for CFG models should therefore be interpreted as intermediate checkpoints rather than final solutions.

**Effect of Guidance Scale.** At 300 epochs, the model’s performance shows a clear preference for stronger guidance: **cfg\_w1** ( $w = 1.5$ ) achieves a poor FID of 33.36, significantly underperforming relative to **cfg\_w2** ( $w = 3.0$ ) and **cfg\_w3** ( $w = 5.0$ ). This indicates that, in the early stages of training, a high guidance scale is essential to compensate for the model’s immature internal representations of artistic styles. At 500 epochs, this benefit persists and is further amplified, as increasing  $w$  from 3.0 to 5.0 improves FID from  $12.79 \pm 0.24$  to  $12.29 \pm 0.31$  and KID from 0.00632 to 0.00378. These results confirm that for ArtBench-10, stronger guidance consistently enhances both fidelity and distributional alignment, with the most significant gains occurring as the model reaches higher states of convergence.

**Effect of Model Capacity.** Reducing the base channel width to  $C = 64$  degrades FID to  $15.19 \pm 0.24$  and KID to  $0.00500 \pm 0.00041$  relative to  $C = 128$ , indicating that the smaller model lacks sufficient capacity to capture the texture diversity of ArtBench-10. The  $C = 128$  configuration provides the best trade-off between model size and generation quality within the evaluated range.

**EMA Ablation.** The EMA and no-EMA variants achieve nearly identical FID (12.29 vs. 12.22), but EMA yields a consistently lower KID (0.00378 vs.



0.00483) and smaller standard deviation across seeds. This suggests that EMA does not prevent the model from reaching a competitive optimum, but stabilises the weight trajectory and reduces sensitivity to late-training stochastic updates, producing more consistent sample quality across random seeds.

**Final Model and Effect of DDIM Steps.** Retraining the best model ( $C = 128$ , with EMA and CFG at  $w = 5.0$ ) on the full dataset yields  $\text{FID} = 9.67 \pm 0.25$  and  $\text{KID} = 0.00406 \pm 0.00049$  when sampled with 20 DDIM steps, a substantial improvement over the 20% subset results.

Crucially, increasing the number of DDIM sampling steps from 20 to 50 further refines the generation quality, pushing the FID down to  $9.28 \pm 0.10$  and the KID to  $0.00264 \pm 0.00031$ . This improvement, accompanied by notably tighter standard deviations, demonstrates that increasing the sampling trajectory depth allows the model to better resolve fine details and achieve higher distributional consistency. The remaining gap relative to the Real vs. Real upper bound ( $\text{FID} = 2.51 \pm 0.03$ ) reflects the inherent difficulty of modelling the full stylistic diversity of ArtBench-10 at  $32 \times 32$  resolution.

## 6 Comparative Analysis

### 6.1 Qualitative Evaluation and Style Diversity

Table 4 reveals a clear quality-cost trade-off across model families. The  $\beta$ -VAE and DCGAN generate 5 000 samples in under 7 seconds, making them suitable for real-time applications. The DCGAN in particular offers a strong cost-to-quality ratio, achieving  $\text{FID} = 18.79$  at near-instant inference. The DDPM requires significantly more computation due to its iterative denoising process, but this cost translates directly into superior quality. Notably, the full dataset DDPM with only 20 DDIM steps already achieves  $\text{FID} = 9.67$  in approximately 17 minutes, surpassing the 20% subset result, while increasing to 50 steps further refines quality to  $\text{FID} = 9.28 \pm 0.10$  at the cost of 42 minutes. This demonstrates that DDIM step count offers a flexible quality-speed trade-off within the diffusion framework, allowing practitioners to balance generation quality against available computational budget.

**Table 4.** Best configuration from each model family with inference time for 5 000 samples.

| Model               | FID ↓                             | KID ↓                                   | Inference <sup>†</sup> |
|---------------------|-----------------------------------|-----------------------------------------|------------------------|
| $\beta$ -VAE        | $97.39 \pm 0.38$                  | $0.10608 \pm 0.00123$                   | ~1.6 s                 |
| DCGAN               | $18.79 \pm 0.36$                  | $0.01201 \pm 0.00053$                   | ~6.7 s                 |
| DDPM (20%)          | $12.29 \pm 0.31$                  | $0.00378 \pm 0.00044$                   | ~17 min                |
| DDPM (full-20steps) | $9.67 \pm 0.25$                   | $0.00406 \pm 0.00049$                   | ~17 min                |
| DDPM (full-50steps) | <b><math>9.28 \pm 0.10</math></b> | <b><math>0.00264 \pm 0.00031</math></b> | ~42 min                |
| Real vs. Real       | $2.51 \pm 0.03$                   | $-0.00003 \pm 0.00020$                  | —                      |

<sup>†</sup>All inference times measured on the same hardware under identical conditions.

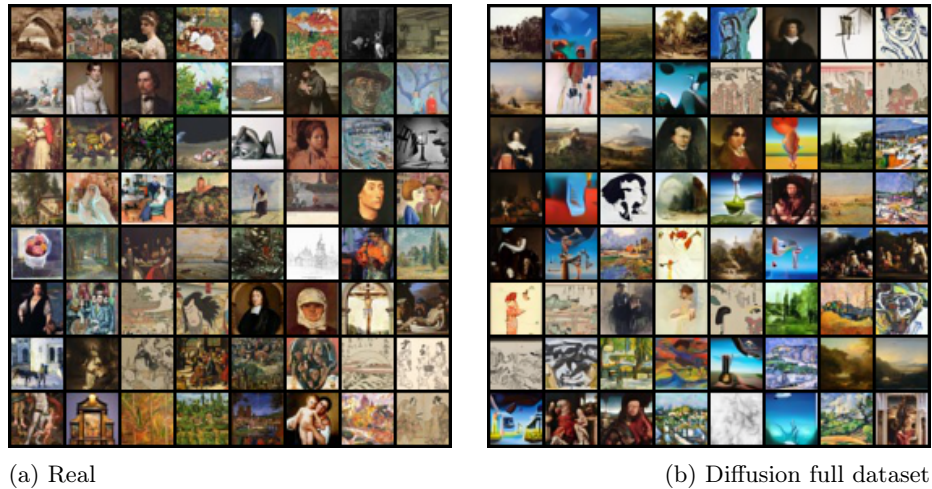
A side-by-side visual comparison across the three model families reveals distinct generative behaviours consistent with their quantitative results (Fig. 1). The  $\beta$ -VAE exhibits severe blurring, capturing broad colour distributions but failing to resolve artistic structure or fine texture, a direct consequence of the MSE reconstruction objective averaging pixel values. The DCGAN achieves significant improvements in sharpness and stylistic contrast, though artistic elements frequently appear fractured, suggesting the model prioritises local textures over global compositional coherence. The DDPM produces the most compelling results, with superior structural integrity, maintaining class-conditional diversity without mode collapse.



**Fig. 1.** Qualitative comparison of the best configuration from each model family. (a) Real ArtBench-10 samples. (b)  $\beta$ -VAE (`vae_b01_z64`, FID = 97.39). (c) DCGAN (`drgan_ls_lr2_ngf256`, FID = 18.79). (d) DDPM (`cfg_w3`, 20% subset, FID = 12.29).

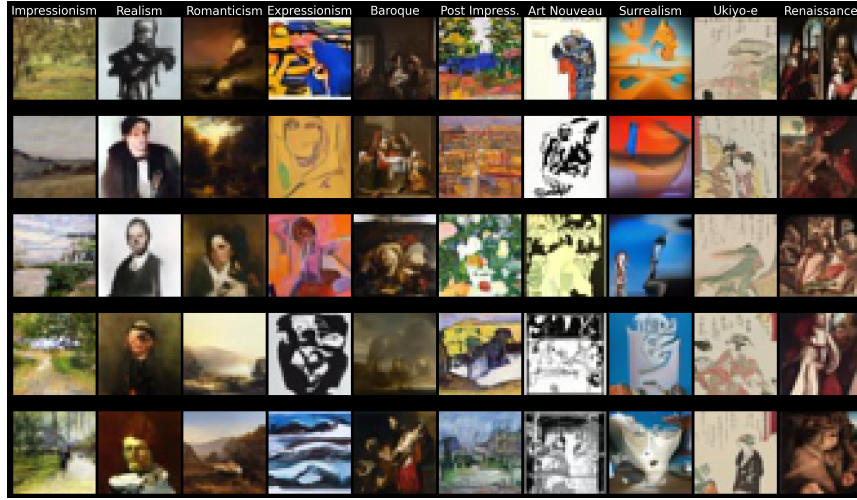
## 6.2 Final Model Performance and Scaling

Retraining the best DDPM on the full dataset with 50 DDIM steps yields  $\text{FID} = 9.28 \pm 0.10$  and  $\text{KID} = 0.00264 \pm 0.00031$ , surpassing all unconditional diffusion baselines in the original ArtBench study [10], including DDIM ( $\text{FID} = 17.56$ ) and Improved DDPM ( $\text{FID} = 15.31$ ). As illustrated in Fig. 2, the model generates visually compelling samples across the full stylistic range of ArtBench-10, substantially closing the gap to the Real vs. Real bound ( $\text{FID} = 2.51$ ).



**Fig. 2.** Best DDPM retrained on the full dataset ( $\text{FID} = 9.28$ ) compared against real ArtBench-10 samples.

Figure 3 shows five conditional samples per artistic style. Impressionism produces soft, hazy landscapes; Realism generates faithful portraits with naturalistic tones; Romanticism is dominated by warm oranges evoking sunsets; Expressionism yields distorted, semi-abstract forms, with one sample resembling Art Nouveau ornamental forms; Baroque generates dark interior scenes including one reminiscent of a battle scene; Post-Impressionism shows vivid structured compositions with a notable coastal city in the second row; Art Nouveau produces nature-inspired forms, with three samples rendered in high-contrast black and white; Surrealism generates abstract imagery with the first sample bearing a striking resemblance to Dalí’s *The Persistence of Memory* [2]; Ukiyo-e captures Japanese visual culture with attempts at kanji script; and Renaissance yields richly coloured religious and royal compositions. The consistency across rows confirms that CFG reliably steers generation towards the target style.



**Fig. 3.** Conditional samples from the best DDPM (full dataset,  $w = 5.0$ ), 5 per artistic style.

## 7 Conclusion

This work presented a systematic evaluation of three generative model families,  $\beta$ -VAEs, DCGANs, and DDPMs, on the ArtBench-10 dataset under a rigorous benchmarking protocol. Results establish a clear performance hierarchy: DDPMs achieve superior sample quality and distributional fidelity, DCGANs offer a competitive middle ground with significantly lower computational cost, and  $\beta$ -VAEs are limited by their MSE reconstruction objective.

For DDPMs, CFG ( $w = 5.0$ ), EMA, and 500 training epochs proved essential, reducing FID from 22.08 (unconditional baseline) to 12.29 in the 20% subset. Scaling to the full dataset with 50 DDIM steps further improved FID to  $9.28 \pm 0.10$ , surpassing all unconditional diffusion baselines reported in the original ArtBench study [10], including Improved DDPM (FID = 15.31) and DDIM (FID = 17.56). Crucially, the conditional generation capabilities enabled by CFG allowed the model to produce diverse, style-consistent samples across all ten artistic classes without mode collapse, as evidenced by the per-class qualitative analysis. For DCGANs, combining label smoothing, asymmetric learning rates, and increasing generator capacity to  $ngf = 256$  reduced FID from 22.42 to 18.79 in under 7 seconds of inference, making it the most practical choice for latency-sensitive applications. For  $\beta$ -VAEs, lower KL regularisation ( $\beta = 0.1$ ) and a smaller latent space ( $d_z = 64$ ) yielded the best FID of 97.39, though all configurations remained substantially behind the other families.

These results highlight the fundamental trade-off between computational cost and generative fidelity: DDPMs set the current standard for high-quality artistic synthesis but require up to 42 minutes to generate 5 000 samples, while DCGANs and VAEs remain orders of magnitude faster at the cost of visual quality.

## References

1. Bińkowski, M., Sutherland, D.J., Arbel, M., Gretton, A.: Demystifying mmd gans (2021), <https://arxiv.org/abs/1801.01401>
2. Dalí, S.: The persistence of memory (1931), oil on canvas, Museum of Modern Art, New York
3. Goodfellow, I.J., Pouget-Abadie, J., Mirza, M., Xu, B., Warde-Farley, D., Ozair, S., Courville, A., Bengio, Y.: Generative adversarial networks (2014), <https://arxiv.org/abs/1406.2661>
4. Heusel, M., Ramsauer, H., Unterthiner, T., Nessler, B., Hochreiter, S.: Gans trained by a two time-scale update rule converge to a local nash equilibrium (2018), <https://arxiv.org/abs/1706.08500>
5. Higgins, I., Matthey, L., Pal, A., Burgess, C., Glorot, X., Botvinick, M., Mohamed, S., Lerchner, A.: beta-VAE: Learning basic visual concepts with a constrained variational framework. In: International Conference on Learning Representations (2017), <https://openreview.net/forum?id=Sy2fzU9gl>
6. Ho, J., Jain, A., Abbeel, P.: Denoising diffusion probabilistic models (2020), <https://arxiv.org/abs/2006.11239>
7. Ho, J., Salimans, T.: Classifier-free diffusion guidance (2022), <https://arxiv.org/abs/2207.12598>
8. Kingma, D.P., Ba, J.: Adam: A method for stochastic optimization (2015), <https://arxiv.org/abs/1412.6980>
9. Kingma, D.P., Welling, M.: Auto-encoding variational bayes (2022), <https://arxiv.org/abs/1312.6114>
10. Liao, P., Li, X., Liu, X., Keutzer, K.: The artbench dataset: Benchmarking generative models with artworks (2022), <https://arxiv.org/abs/2206.11404>
11. Nichol, A., Dhariwal, P.: Improved denoising diffusion probabilistic models (2021), <https://arxiv.org/abs/2102.09672>
12. Radford, A., Metz, L., Chintala, S.: Unsupervised representation learning with deep convolutional generative adversarial networks (2016), <https://arxiv.org/abs/1511.06434>
13. Salimans, T., Goodfellow, I., Zaremba, W., Cheung, V., Radford, A., Chen, X.: Improved techniques for training gans (2016), <https://arxiv.org/abs/1606.03498>
14. Song, J., Meng, C., Ermon, S.: Denoising diffusion implicit models (2022), <https://arxiv.org/abs/2010.02502>